MAHATMA GANDHI MEMORIAL COLLEGE

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Mini Project Report On

"Plant leaf disease detection using deep learning"

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MAHATMA GANDHI MEMORIAL COLLEGE
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POST GRADUATION
DEPARTMENT OF COMPUTER SCIENCE

CERTIFICATE

This is to certify that the mini project entitled **"**Plant leaf disease detection using deep learning**"** has been carried out by **Sumanth (Reg No: P05MG24S038023)**, student of **III semester M.Sc(Computer Science)** under the guidance of **Ms. Pallavi,** Assistant Professor, PG Department of Computer Science, Mahatma Gandhi Memorial College, Udupi. This dissertation is submitted in partial fulfilment of the requirement for the award of **Master of Science (Computer Science)** of **Mangalore University** during the Academic year **2025–2026**.

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ABSTRACT

Plant diseases significantly affect agricultural productivity and food security if not identified at an early stage. Traditional disease detection methods rely on manual inspection, which is time-consuming, labor-intensive, and prone to errors. To overcome these limitations, this project presents an automated plant leaf disease detection system using deep learning techniques.Convolutional Neural Network (CNN)–based models, namely EfficientNetB0 and MobileNet, are implemented and evaluated using the PlantVillage dataset containing healthy and diseased leaf images. Image preprocessing and data augmentation techniques are applied to improve model generalization. Model performance is assessed using accuracy, precision, recall, and F1-score metrics. Experimental results show that EfficientNetB0 achieves high accuracy of about 99%, while MobileNet attains around 93% accuracy with lower computational requirements. The trained model is integrated into a web-based application that enables users to upload leaf images and obtain disease predictions. The proposed system provides an efficient and accurate solution for early plant disease detection and supports the use of deep learning in precision agriculture.

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1.Introduction

1.1 Introduction

Plant diseases are a major challenge in agriculture, causing significant crop losses and affecting food security worldwide. Traditional methods of disease detection rely on manual inspection, which is time-consuming, labor-intensive, and prone to human error. With the advancements in computer vision and deep learning, automated detection systems have emerged as effective tools for early identification of plant diseases. Deep learning models, particularly Convolutional Neural Networks (CNNs), have shown high accuracy in classifying plant leaf images and detecting disease patterns. This project aims to study and implement deep learning-based plant disease detection methods to improve accuracy and efficiency in identifying plant diseases.Deep learning–based approaches have emerged as highly effective solutions for plant disease detection, significantly outperforming traditional methods in terms of accuracy and robustness. Advanced models combining convolutional neural networks with modern architectures demonstrate strong generalization capabilities while reducing dependence on large labeled datasets, making them suitable for scalable agricultural applications. Studies show that deep learning techniques can successfully extract complex features from plant leaf images and achieve high classification accuracy, often exceeding 95%. However, challenges such as noisy backgrounds, varying illumination, dataset imbalance, computational complexity, and real-time deployment constraints still remain. Overall, these findings highlight the potential of deep learning in precision agriculture while emphasizing the need for efficient, accurate, and deployable models for real-world plant disease detection systems.

1.2 Problem statement

Manual detection of plant diseases is inefficient, error-prone, and impractical for large-scale farming. Farmers face significant challenges in real-time crop monitoring, particularly when managing thousands of plants spread across large agricultural fields. Existing manual methods are unable to process large volumes of plant image data efficiently, which leads to delays in early disease identification. Therefore, there is a strong need for an automated, real-time plant disease detection system capable of analyzing large datasets of leaf images, accurately classifying diseases, and providing timely alerts to farmers. Such a system can help reduce crop losses and significantly improve agricultural productivity.

1.3 Scope

The scope of this project is to develop an efficient and user-friendly web-based plant disease detection system using machine learning and deep learning techniques. The project focuses on early identification of various plant diseases through leaf image analysis using CNN-based models, supported by image preprocessing and data augmentation to improve accuracy and generalization. Publicly available datasets are used for training and evaluation, and model performance is assessed using standard metrics such as accuracy, precision, recall, and F1-score. The developed web application allows users to upload plant leaf images and receive accurate disease predictions, reducing dependency on manual inspection. The system also provides a foundation for future enhancements such as real-time detection, scalability to multiple crops and diseases, and integration with smart and precision agriculture technologies to improve overall agricultural productivity.

2. Literature review

Several researchers have investigated the application of deep learning techniques for automated plant disease detection using leaf images, with a strong focus on improving accuracy and robustness under real-world conditions. Mindhe examined the effectiveness of Convolutional Neural Networks (CNNs) for identifying plant diseases from leaf images, emphasizing preprocessing techniques, data augmentation, and model optimization while addressing challenges such as illumination variations, complex backgrounds, and similarity between disease classes [1]. Liu presented a comprehensive review of deep learning methods for both plant disease and pest detection, comparing classification, detection, and segmentation networks in terms of accuracy and computational efficiency, and highlighting limitations encountered during practical agricultural deployment [2]. Pacal conducted a large-scale systematic review of 160 studies on plant disease detection, categorizing existing research into classification, detection, and segmentation tasks, and identifying commonly used datasets, network architectures, and existing research gaps [3]. Shoaib reviewed recent deep learning models for plant disease detection, with particular emphasis on transfer learning, pre-trained CNNs, and hybrid architectures, and discussed issues related to model evaluation and cross-crop generalization [4].

Dhaka surveyed multiple CNN-based architectures for plant leaf disease classification, analyzing preprocessing, data augmentation, and optimization strategies, while also highlighting accuracy improvements and limitations when models are applied outside controlled environments [5]. Nyawose reviewed both traditional machine learning and deep learning approaches for plant disease detection, comparing their performance and identifying key challenges such as dataset imbalance and reduced effectiveness in real-field scenarios [6]. Roumeliotis explored the integration of multimodal Large Language Models with CNNs for plant disease classification, demonstrating that fine-tuned multimodal frameworks can enhance accuracy and generalization while reducing dependency on large labeled datasets [7].

Upadhyay provided a comprehensive review of deep learning and computer vision techniques used in plant disease detection, covering datasets, preprocessing strategies, modern architectures, and challenges related to real-world agricultural implementation [8]. Shoaib further reviewed deep learning approaches for both plant disease and pest detection, reporting high accuracy for CNN-based models while identifying issues such as background noise, lighting variability, and constraints in real-time inference [9]. Khan reviewed automated plant disease detection systems that combine machine learning, deep learning, and remote sensing technologies, emphasizing challenges related to model generalization, dataset diversity, and computational limitations in practical applications [10].

Overall, these studies demonstrate that CNN-based deep learning models and image-based analysis are widely used for plant disease detection and achieve high accuracy under controlled conditions. However, many existing works remain limited to experimental evaluations and review-based analyses, with fewer implementations of complete, deployable systems for real agricultural environments. Most approaches rely heavily on image datasets captured under ideal conditions, and deep learning models often require large, well-labeled datasets and extensive tuning to maintain performance in diverse field conditions. These limitations highlight the need for more robust, scalable, and practical plant disease detection systems that can generalize effectively to real-world agricultural settings.

3. Objectives

- To review and analyze the latest deep learning techniques forplant disease detection.
- To implement a deep learning model capable of detecting plant diseases from leaf images.
- To compare different deep learning architectures and identify the most effective approach.

4. Methodology

4.1 Software and hardware specification

The project is developed using Python as the primary programming language, with TensorFlow and Keras libraries for implementing deep learning models such as MobileNetV2 and EfficientNetB0. Image preprocessing and data augmentation are performed using the ImageDataGenerator module, and the PlantVillage dataset is used with an 80% training and 20% validation and testing split. Model development and experimentation are carried out using Jupyter Notebook, while Visual Studio Code is used for application development, and Flask is used to create a web-based interface. The system is executed on a standard computer system without GPU support, making the implementation suitable for CPU-based environments.

Workflow:

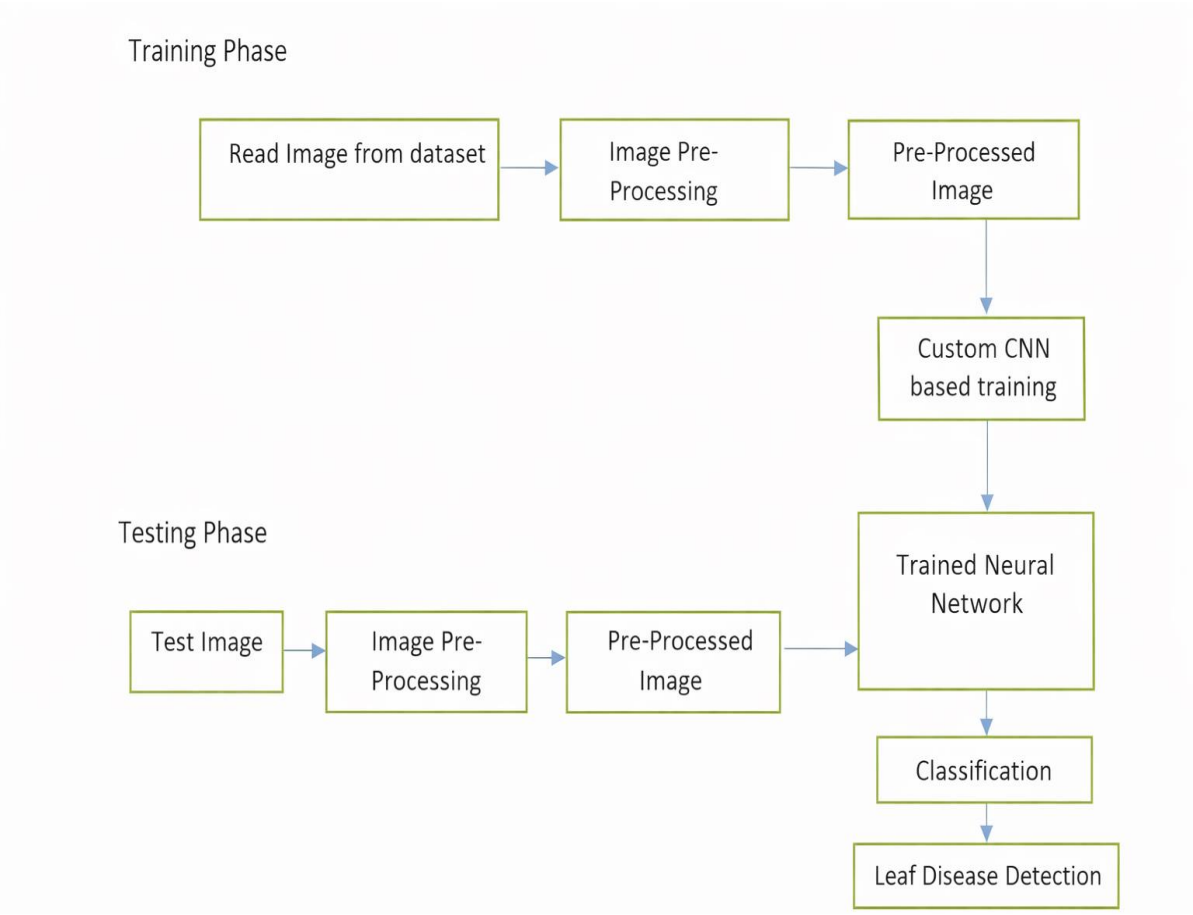


Fig 4.1 Block Diagram

4.2 Dataset and Data Splitting

The PlantVillage dataset is used in this project, which contains plant leaf images categorized into 15 different classes, including both healthy and diseased plants. This dataset provides a diverse and well-labeled collection of images suitable for training deep learning models. To ensure proper model training and evaluation, the dataset is divided into training, validation, and testing subsets. Out of the total dataset, 7,564 images are used for training the model, while 946 images are used for validation and another 946 images are used for testing. This structured data splitting helps in monitoring model performance during training and ensures reliable evaluation on unseen data.

4.3 Image Acquistion

Image acquisition refers to the process of obtaining plant leaf images required for building and evaluating the disease detection system. In this project, images are collected from the PlantVillage dataset, which provides a large collection of labeled leaf images representing various plant diseases and healthy plant conditions. The dataset contains images captured under controlled environments with consistent lighting and background, ensuring image clarity and quality. These images are stored in standard digital formats and serve as the input for preprocessing and deep learning model training.

4.4 Image Preprocessing

Image preprocessing is performed to improve the quality and consistency of plant leaf images before training the models. All images are resized to a fixed size of 224 × 224 pixels to ensure uniform input across different deep learning architectures. Data augmentation techniques such as rotation, width and height shifting, zooming, and horizontal flipping are applied to increase dataset variability and reduce overfitting. These preprocessing steps help the models learn robust features and improve classification performance.

4.5 Training process

The PlantVillage dataset is divided into training, validation, and testing sets, with approximately 80% of the images used for training and 20% used for validation and testing. All images are resized to 224 × 224 pixels, and data augmentation is applied to the training set to improve generalization. The models are trained with a batch size of 16. The EfficientNet-based model is trained for up to 30 epochs with early stopping applied at the 23rd epoch, while the MobileNet-based model is trained for 25 epochs. Validation data is used during training to monitor performance, and the best-trained model is saved for final testing and evaluation.

4.6 Validation

Validation is performed to evaluate the model's performance on unseen data during training and to prevent overfitting. In this project, **946 images**, which constitute approximately **10% of the total dataset**, are used for validation. These images are not included in the training process and are used to monitor the model's learning behaviour after each epoch. Validation accuracy and loss are observed during training, and early stopping is applied when no further improvement is detected, ensuring better model generalization.

Evaluation Metrics:

- **Accuracy:** Measures the overall correctness of the model by calculating the percentage of correctly classified images out of the total images.
- **Precision:** Indicates how many of the predicted disease cases are actually correct.
- **Recall:** Measures the ability of the model to correctly identify all actual disease cases.
- **F1-Score:** The harmonic mean of precision and recall, providing a balanced measure of model performance.
- **Support:** number of samples belonging to each class in the dataset.

4.7 Testing phase

The testing phase is performed to evaluate the final performance of the trained models on completely unseen data. In this project, 946 images, accounting for approximately 10% of the PlantVillage dataset, are used for testing. These images are not involved in either training or validation.

4.8 Final Evaluation

The final evaluation of the plant disease detection system is conducted using the test dataset consisting of 946 images from the PlantVillage dataset, which were not used during training or validation. This phase assesses the overall performance and generalization capability of the trained CNN-based models. The test images are passed through the complete processing pipeline, including image preprocessing, CNN-based feature extraction, and disease classification.

In addition to dataset-based testing, the trained model is integrated into a web-based interface, where users can upload plant leaf images. The uploaded image is processed by the trained model, and the predicted disease name is displayed to the user. The model's performance is evaluated using standard metrics such as accuracy, precision, recall, F1-score, and support. The results demonstrate that the system effectively classifies plant diseases and provides reliable predictions, confirming its suitability for real-world plant disease detection applications.

Flow Diagram

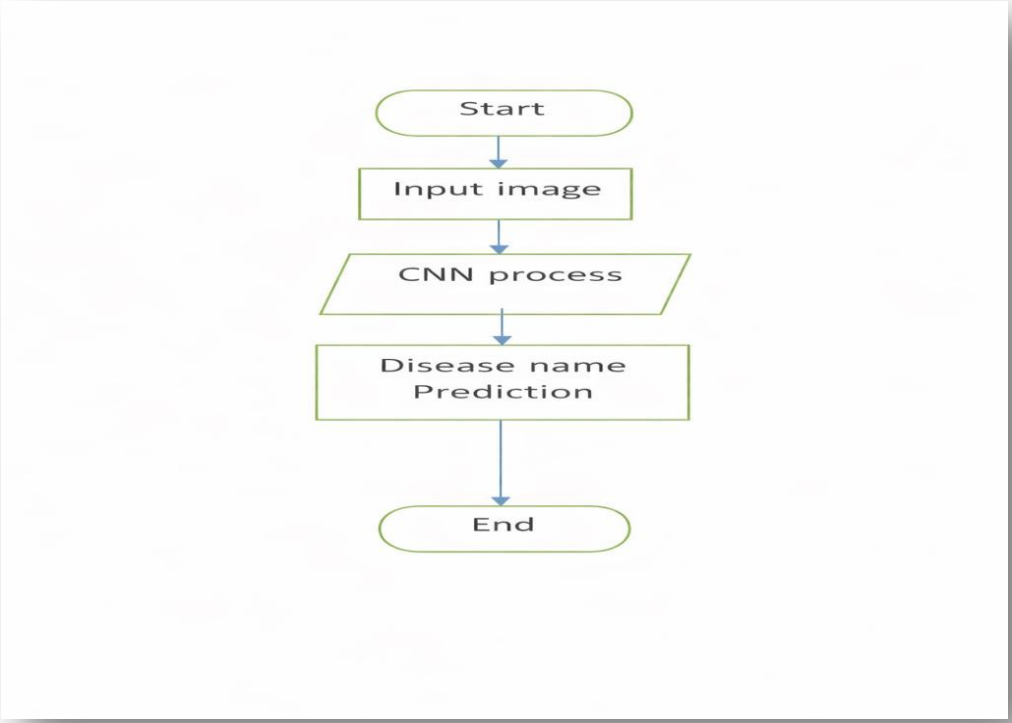
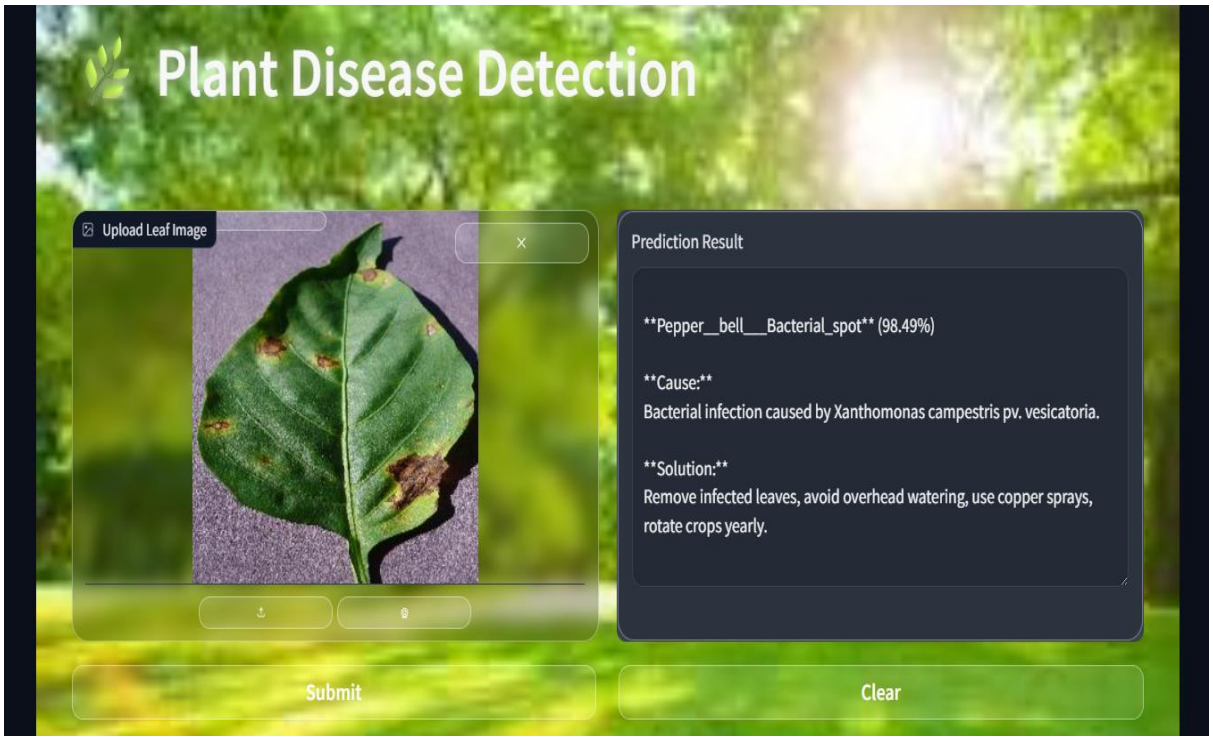


Fig 4.2 Flow Diagram

4.9. Web Application

A web-based application was developed using **Gradio** to provide an interactive and user-friendly interface for plant disease detection. The application allows users to upload plant leaf images and receive disease predictions along with confidence scores. Gradio simplifies the deployment of the trained deep learning model and enables quick testing and demonstration of the system without complex frontend development. This web application serves as an effective platform for showcasing the model's performance and usability.

Web Interface:



4.3 Plant Disease Detection Using Interface

5. Result and Discussion

The performance of the proposed plant disease detection system was evaluated using two CNN-based models, EfficientNet and MobileNet, on the PlantVillage dataset containing 15 plant disease and healthy classes. The evaluation was performed using 946 test images that were not used during training or validation.

5.1 EfficientNet Model Performance

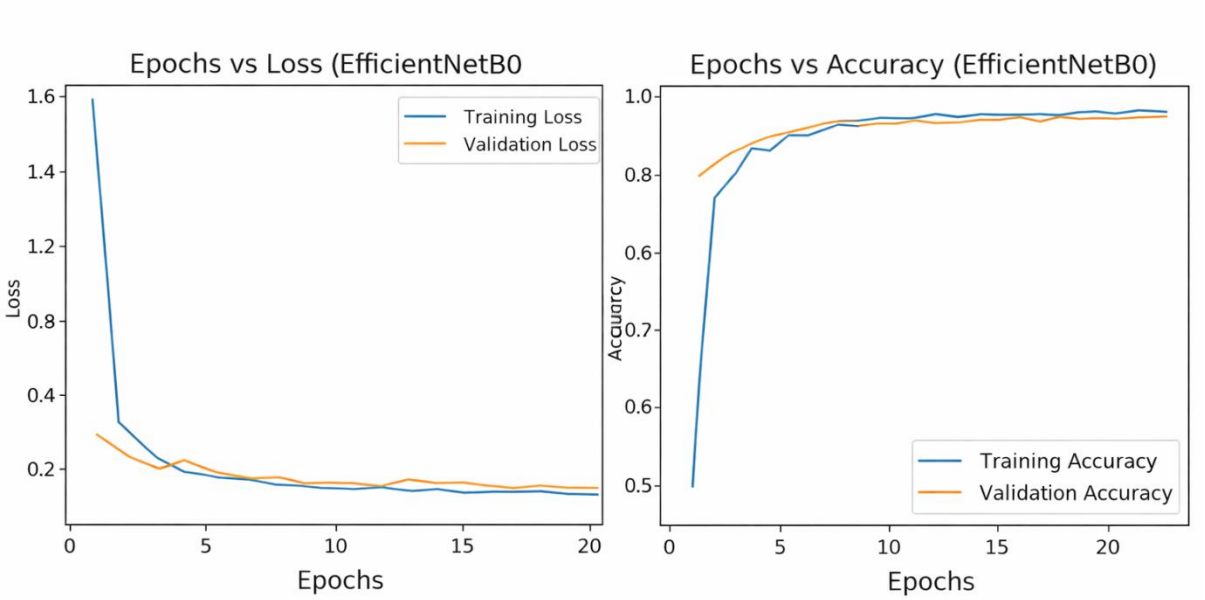


Fig 5.1 Training vs Validation accuracy and loss(EfficientNetB0)

The given Fig 5.1 showed very good performance during training and testing. It was trained for a maximum of 30 epochs, with early stopping applied at the 23rd epoch to avoid overfitting. The training accuracy increased steadily, and the validation accuracy reached about 98.6%, indicating good generalization.

During testing, the EfficientNet model achieved a test accuracy of 99.47%. Most classes achieved high precision, recall, and F1-score values, showing that the model accurately classified different plant diseases and healthy leaves. The low loss values further confirm the reliability of the model.

The Fig 5.2 shows a strong diagonal dominance, indicating that almost all plant disease classes were correctly classified with very minimal misclassification. Only a few samples of visually similar tomato virus classes show slight confusion, while most classes achieve near-perfect prediction counts. This confirms the model's high robustness and its ability to clearly distinguish between different plant diseases.

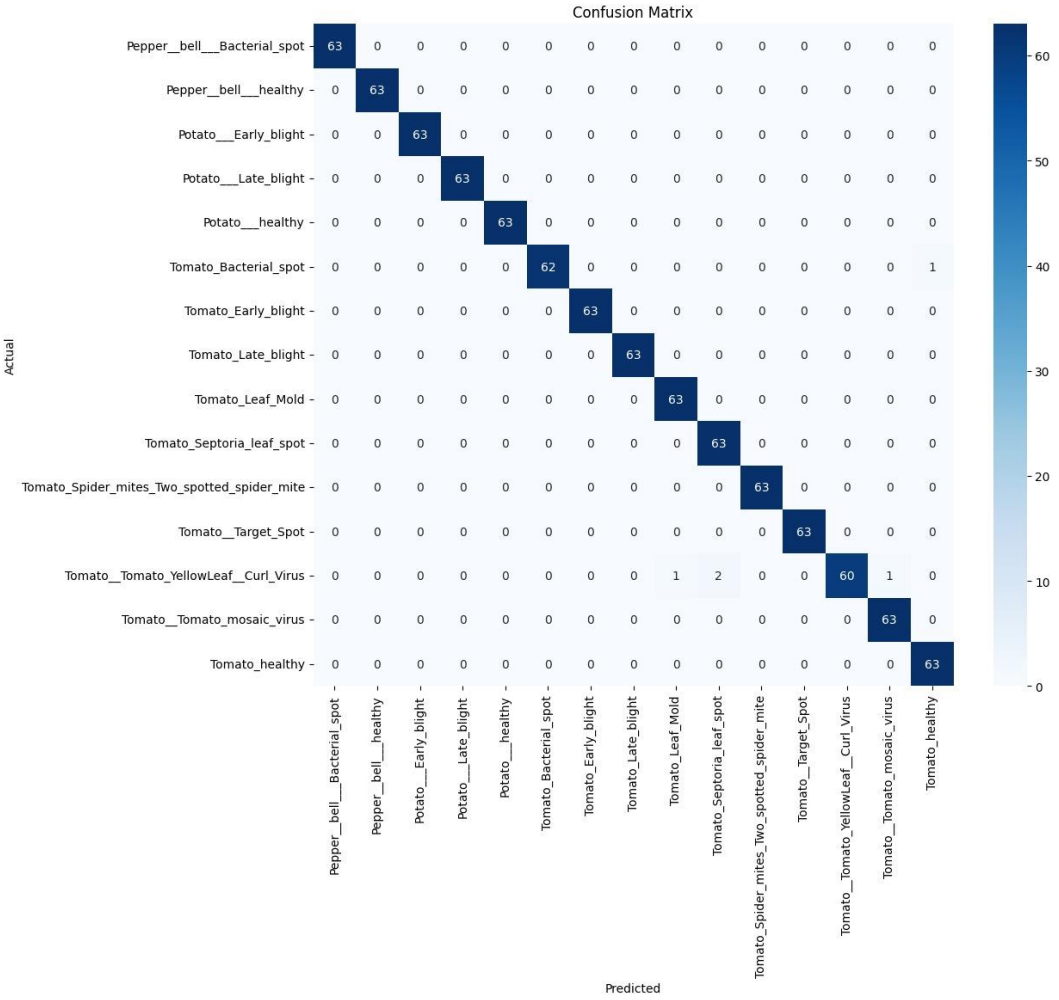


Fig 5.2 Confusion Matrix of EfficientNetB0

5.2 MobileNet Model Performance

The MobileNet model was trained for 25 epochs with a batch size of 16. The Fig 5.3 showed gradual improvement in accuracy during training. The final validation accuracy reached around 89–90%, indicating stable learning.

On the test dataset, MobileNet achieved a test accuracy of 93.13%. Most classes showed good precision and recall, though a few disease classes had slightly lower performance. This is mainly due to the lightweight nature of the MobileNet architecture.

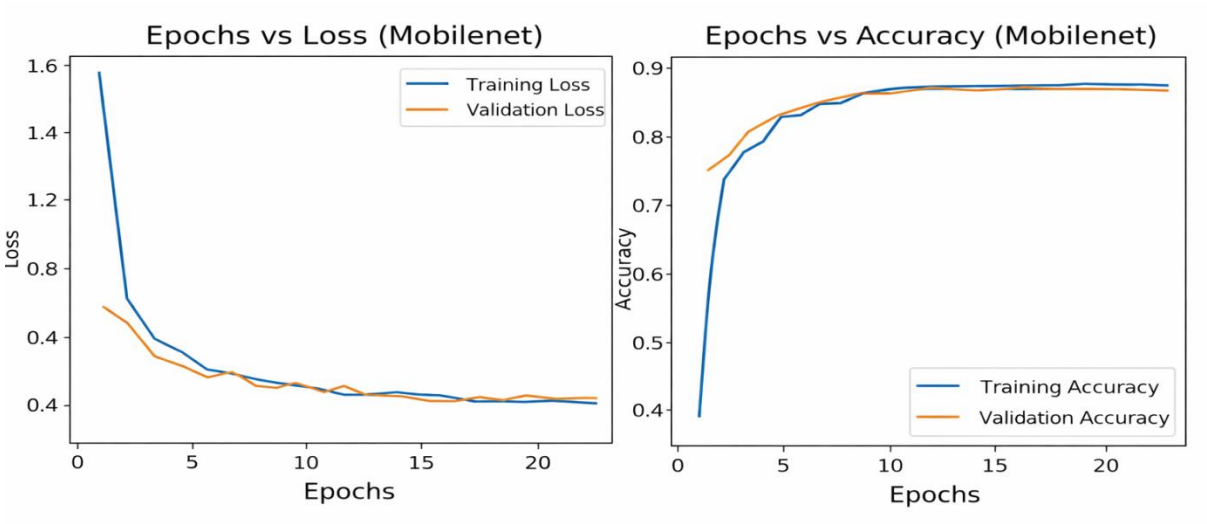


Fig 5.3 Training vs Validation accuracy and loss(Mobilenet)

In contrast, the MobileNetV2 confusion matrix shows more off-diagonal values, particularly among tomato disease classes such as early blight, late blight, and leaf mold, indicating higher misclassification. Although MobileNet performs well for several classes, its lightweight architecture leads to reduced discrimination between diseases with similar visual patterns. This explains the lower overall accuracy of MobileNet compared to EfficientNet.

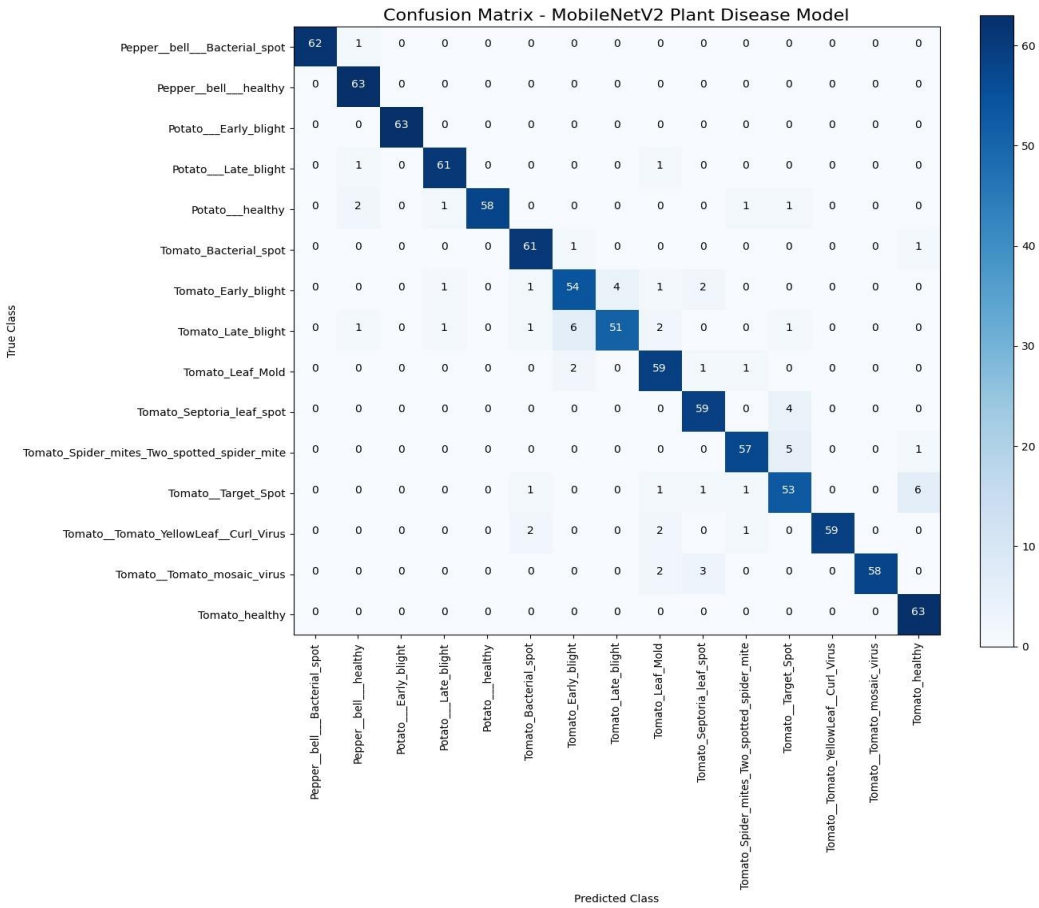


Fig 5.2 Confusion Matrix of MobileNet

5.3 Comparative Discussion

The results show that EfficientNet performs better than MobileNet in terms of accuracy and class-wise performance. EfficientNet is able to learn complex disease features more effectively, but it requires more training time.MobileNet, in contrast, is faster and requires less computation, making it suitable for systems with limited resources. Although its accuracy is lower than EfficientNet, it still provides reliable results.

5.4 Model Evaluation Metrics

The EfficientNet model achieved an excellent overall accuracy of 99%, demonstrating consistently high precision, recall, and F1-scores across all plant disease classes, which indicates strong learning and generalization capability. In comparison, the MobileNet model obtained an accuracy of 93%, showing slightly lower performance for certain disease classes due to its lightweight architecture. Overall, EfficientNet outperformed MobileNet and proved to be more reliable for accurate plant disease classification.

Table 5.1 Classification Report-MobileNet

Class / Metric	Precision	Recall	F1-score	Support
Pepper__bell___Bacterial_spot	1.00	0.98	0.99	63
Pepper__bell___healthy	0.93	1.00	0.96	63
Potato___Early_blight	1.00	1.00	1.00	63
Potato___Late_blight	0.95	0.97	0.96	63
Potato___healthy	1.00	0.92	0.96	63
Tomato_Bacterial_spot	0.92	0.97	0.95	63
Tomato_Early_blight	0.86	0.86	0.86	63
Tomato_Late_blight	0.93	0.81	0.86	63
Tomato_Leaf_Mold	0.87	0.94	0.90	63
Tomato_Septoria_leaf_spot	0.89	0.94	0.91	63
Tomato_Spider_mites_Two_spotted_spider_mite	0.93	0.90	0.92	63
Tomato_Target_Spot	0.83	0.84	0.83	63
Tomato_Tomato_YellowLeaf_Curl_Virus	1.00	0.92	0.96	64
Tomato_Tomato_mosaic_virus	1.00	0.92	0.96	63
Tomato_healthy	0.89	1.00	0.94	63
Accuracy	0.93	—	—	946
Macro Average	0.93	0.93	0.93	946
Weighted Average	0.93	0.93	0.93	946

Table 5.2 Classification Report-EfficientNetB0

Class / Metric	Precisio	Recall	F1-scsore	Suppor t
Pepper__bell___Bacterial_spot	1.00	1.00	1.00	63
Pepper__bell___healthy	1.00	1.00	1.00	63
Potato___Early_blight	1.00	1.00	1.00	63
Potato___Late_blight	1.00	1.00	1.00	63
Potato___healthy	1.00	1.00	1.00	63
Tomato_Bacterial_spot	1.00	0.98	0.99	63
Tomato_Early_blight	1.00	1.00	1.00	63
Tomato_Late_blight	1.00	1.00	1.00	63
Tomato_Leaf_Mold	0.98	1.00	0.99	63
Tomato_Septoria_leaf_spot	0.97	1.00	0.98	63
Tomato_Spider_mites_Two_spotted_spider_mite	1.00	1.00	1.00	63
Tomato_Target_Spot	1.00	1.00	1.00	63
Tomato_Tomato_YellowLeaf_Curl_Virus	1.00	0.94	0.97	64
Tomato_Tomato_mosaic_virus	0.98	1.00	0.99	63
Tomato_healthy	0.98	1.00	0.99	63
Accuracy	0.99	—	—	946
Macro Average	0.99	0.99	0.99	946
Weighted Average	0.99	0.99	0.99	946

5.5 Overall Discussion

The results confirm that CNN-based models are effective for plant disease detection. EfficientNet is suitable for applications requiring high accuracy, while MobileNet offers a good balance between accuracy and efficiency. The use of a web-based interface further shows the practical usefulness of the proposed system in real-world agricultural applications.

6. Conclusion and Future Scope

6.1 Conclusion

This project successfully developed an automated plant disease detection system using deep learning techniques. CNN-based models were trained and evaluated using the PlantVillage dataset containing multiple plant disease and healthy classes. The results demonstrate that deep learning methods are highly effective in accurately identifying plant diseases from leaf images.Among the evaluated models, EfficientNet achieved very high accuracy and showed strong classification performance, while MobileNet provided a good balance between accuracy and computational efficiency. These results highlight the strengths of different CNN architectures for plant disease detection.

Overall, the proposed system reduces the dependence on manual disease inspection and supports early and accurate detection of plant diseases, thereby contributing to improved agricultural productivity and decision-making.

6.2 Limitations

Although the proposed plant disease detection system demonstrates high accuracy on the PlantVillage dataset, certain limitations are observed when applied to real-world scenarios. The model shows reduced prediction reliability on images captured from external sources due to variations in lighting conditions, background clutter, and image quality. Moreover, the system is not fully optimized for real-time disease detection using a webcam, which can impact both prediction accuracy and performance. These observations indicate the need for further enhancements to improve robustness and adaptability to real-world image variations.

6.3 Future Scope

The future scope of this project includes training the model with a larger and more diverse set of plant leaf images captured under real-world conditions. This will help improve prediction accuracy for images taken using low-end devices and varying lighting environments. The system can also be enhanced to support accurate real-time disease detection using a webcam. Additionally, integrating sensor data such as temperature, humidity, and soil moisture can further improve prediction reliability and enable smarter decision-making in precision agriculture.

References

1. Mindhe, O., Kurkute, O., Naxikar, S., & Raje, N. (2020). Plant disease detection using deep learning. International Research Journal of Engineering and Technology, 7(4), 2497–2503.

2. Liu, J., & Wang, X. (2021). Plant diseases and pests detection based on deep learning: A review. Plant Methods, 17(1), 22.

3. Pacal, I., Kunduracioglu, I., Alma, M. H., Deveci, M., Kadry, S., Nedoma, J., Slany, V., & Martinek, R. (2024). A systematic review of deep learning techniques for plant diseases. Artificial Intelligence Review, 57(11), 304.

4. Shoaib, M., Shah, B., El-Sappagh, S., Ali, A., Ullah, A., Alenezi, F., Gechev, T., Hussain, T., & Ali, F. (2023). An advanced deep learning models-based plant disease detection: A review of recent research. Frontiers in Plant Science, 14, 1158933.

5. Dhaka, V. S., Meena, S. V., Rani, G., Sinwar, D., Ijaz, M. F., & Woźniak, M. (2021). A survey of deep convolutional neural networks applied for prediction of plant leaf diseases. Sensors, 21(14), 4749.

6. Nyawose, T., Maswanganyi, R. C., & Khumalo, P. (2025). A review on the detection of plant disease using machine learning and deep learning approaches. Journal of Imaging, 11(10), 326.

7. Roumeliotis, K. I., Sapkota, R., Karkee, M., Tselikas, N. D., & Nasiopoulos, D. K. (2025). Plant disease detection through multimodal large language models and convolutional neural networks. IEEE Transactions on Automation Science and Engineering, Under Review.

8. Upadhyay, A., Chandel, N. S., Singh, K. P., Chakraborty, S. K., Nandede, B. M., Kumar, M., Subeesh, A., Upendar, K., Salem, A., & Elbeltagi, A. (2025). Deep learning and computer vision in plant disease detection: A comprehensive review of techniques, models, and trends in precision agriculture. Artificial Intelligence Review, 58, 92.

9. Shoaib, M., Sadeghi-Niaraki, A., Ali, F., Hussain, I., & Khalid, S. (2025). Leveraging deep learning for plant disease and pest detection: A comprehensive review and future directions. Frontiers in Plant Science, 16, 1538163.

10. Khan, S. U., Alsuhaibani, A., Alabduljabbar, A., Almarshad, F., Altherwy, Y., & Akram, T. (2025). A review on automated plant disease detection: Motivation, limitations, challenges, and recent advancements for future research. Journal of King Saud University – Computer and Information Sciences, 37, 34.